

Bangla Banking Chatbot and Intent Detection Using MuRIL BERT

1. Introduction

With the rapid growth of digital banking, customers expect quick and efficient support services. However, many existing banking chatbots primarily support English, making it difficult for Bangla-speaking users to communicate effectively. This project proposes an AI-powered Bangla Banking Chatbot that understands Bangla and Banglish banking queries through intent detection. The chatbot uses the MuRIL (Multilingual Representations for Indian Languages) BERT model to classify user intents and provide appropriate banking responses. The system aims to improve customer experience, reduce response time, and automate common banking services.

2. Problem Statement

Many banking institutions do not have intelligent Bangla-language chatbots capable of accurately understanding customer requests. Customers often experience delays in obtaining information about account balances, loans, ATM locations, money transfers, and other banking services. Traditional rule-based chatbots cannot effectively understand different writing styles or variations of the Bangla language. Therefore, there is a need for an intelligent NLP-based chatbot that can accurately recognize customer intent and provide instant responses.

3. Objectives

General Objective

To develop an intelligent Bangla Banking Chatbot using the MuRIL BERT model for accurate banking intent detection and automated customer support.

Specific Objectives

- To build a Bangla banking query dataset with labeled intents.
- To preprocess and prepare the dataset for model training.

- To fine-tune the MuRIL BERT model for Bangla intent classification.
 - To classify user queries into predefined banking intents.
 - To generate appropriate banking responses based on detected intents.
 - To evaluate the system using Accuracy, Precision, Recall, and F1-score.
 - To compare the performance of MuRIL BERT with a traditional TF-IDF + Logistic Regression model.
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4. Scope of the Project

The proposed chatbot will support common banking services, including:

- Balance Inquiry
- Loan Information
- ATM Location
- Mobile Banking
- Money Transfer
- Card Issues
- Account Opening
- Cheque Book Request
- Transaction History
- Customer Support

The chatbot is designed to understand both Bangla and Banglish user queries and provide predefined responses based on detected intents.

5. Literature Review

Recent developments in Natural Language Processing (NLP) have significantly improved chatbot performance through transformer-based language models. Models such as BERT and MuRIL are capable of understanding contextual information in multilingual text. MuRIL is specifically designed for Indian subcontinent languages, including Bangla, making it highly suitable for Bangla banking applications. Compared to traditional machine learning techniques such as TF-IDF with Logistic Regression, transformer-based models generally provide higher accuracy and better contextual understanding.

6. Proposed Methodology

Step 1: Data Collection

A Bangla banking dataset containing customer queries and corresponding intent labels will be collected and stored in Excel format.

Step 2: Data Preprocessing

The collected dataset will be cleaned by removing duplicate entries, encoding intent labels, and splitting the data into training and testing sets.

Step 3: Model Development

The Bangla text will be tokenized using the MuRIL tokenizer. The MuRIL BERT model will then be fine-tuned for multiclass intent classification.

Step 4: Intent Prediction

When a user enters a banking query, the trained model predicts the corresponding banking intent.

Step 5: Response Generation

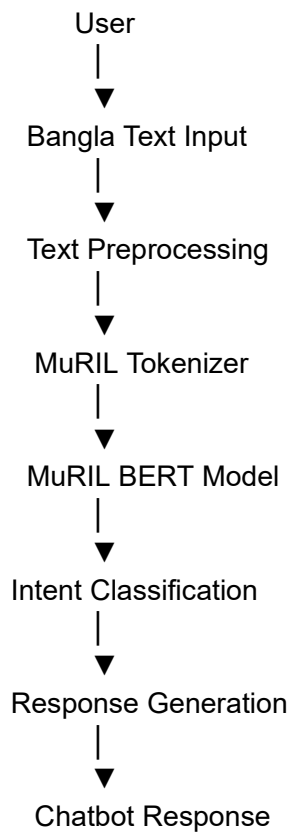
The predicted intent is matched with predefined banking responses, and the chatbot returns the appropriate reply.

Step 6: Performance Evaluation

The model performance will be evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

7. System Architecture



8. Tools and Technologies

Component	Technology
Programming Language	Python
Development Environment	Google Colab
Deep Learning Framework	PyTorch
NLP Library	Hugging Face Transformers
Data Processing	Pandas
Machine Learning	Scikit-learn
Visualization	Matplotlib

Model

MuRIL BERT

9. Expected Outcomes

The proposed system is expected to:

- Accurately detect banking-related intents from Bangla text.
- Provide instant responses to common banking queries.
- Improve customer service efficiency.
- Reduce the workload of banking customer support.
- Achieve high classification accuracy using MuRIL BERT.

10. Performance Evaluation Metrics

The chatbot will be evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics will measure the effectiveness and reliability of the intent classification model.

11. Future Work

Future improvements of the project may include:

- Integration with real banking APIs.
- Voice-based Bangla banking assistant.
- Speech-to-Text and Text-to-Speech support.
- Mobile application deployment.
- Secure customer authentication.
- Generative AI-based conversational responses instead of predefined replies.

12. Conclusion

This project proposes an intelligent Bangla Banking Chatbot using the MuRIL BERT transformer model for banking intent detection. The chatbot is capable of understanding Bangla and Banglish banking queries and providing relevant responses. By automating common banking services, the system can improve customer satisfaction, reduce response time, and enhance digital banking accessibility for Bangla-speaking users. The proposed solution also demonstrates the effectiveness of transformer-based NLP techniques in multilingual banking applications.